

Your
Team
Name

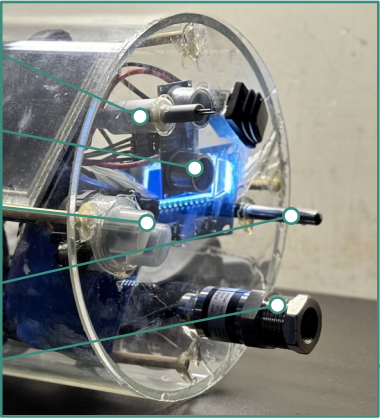
AquaSDR • TECHNICAL APPROACH

Adaptive Software-Defined Sonar Transmitter for AUVs

REAL SENSORS → ON-CHIP LOGIC → REAL PAYLOAD



SENSE • REAL ENVIRONMENT



TDS / SALINITY PROXY

HC-SR04 • AIR RANGE

TURBIDITY

DS18B20 TEMPERATURE

PRESSURE / DEPTH

ADC • 1-Wire • I²C

ADDITIONAL SENSORS



BMP390
cabin
pressure
I²C

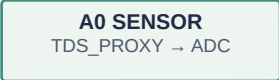


**IMU +
magnetometer**
ISM330 • MMC5983
I²C



INA219 x2
MOUNTED • NOT
COMMISSIONED

ONE ADAPTATION POLICY • TWO INPUTS



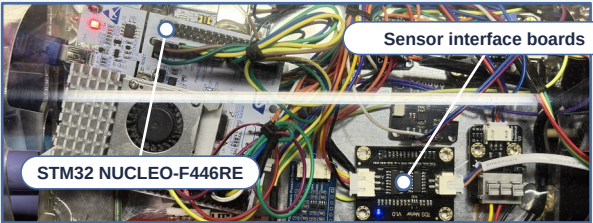
A0 SENSOR
TDS_PROXY → ADC



WEB COMMAND
console → UART

STM32F446RE • EMBEDDED CONTROL methodology & process

THE NEXT PING IS COMPUTED ON THE MCU



STM32 NUCLEO-F446RE

Sensor interface boards

MCU ADC CAPTURE
LFM example • on-board
Centre **42 kHz**
Sweep **4 kHz**
Pulse **2 ms**
Window **Hann**
330/330 cmds ACKed
0 DMA underruns

- 01 ACQUIRE**
ADC • 1-Wire • I²C
stale / out-of-range inputs rejected
- 02 DECIDE**
smoothing filter + hysteresis
CLEAR / TRANSITION / MURKY
- 03 SYNTHESISE**
LFM chirp • geometric sweep • Barker-13
Hann • Hamming • Blackman windows
- 04 STREAM**
TIM6 + DMA → 12-bit DAC1 / PA4
1 MS/s • 0 underruns • CPU free between transfers
- 06 SELF-VALIDATE**
ADC + DMA capture of the DAC output
FFT • spectrogram • matched filtering

built / running

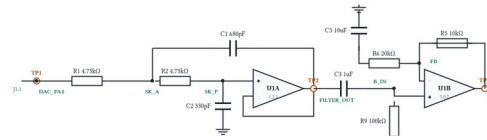
designed / next build

ADAPTED NEXT PING

	CLEAR	TRANSITION	MURKY
Centre	42 kHz	40 kHz	38 kHz
Bandwidth	4 kHz	2 kHz	1 kHz
Pulse	2 ms	4 ms	8 ms
Amplitude	35 %	50 %	62 %

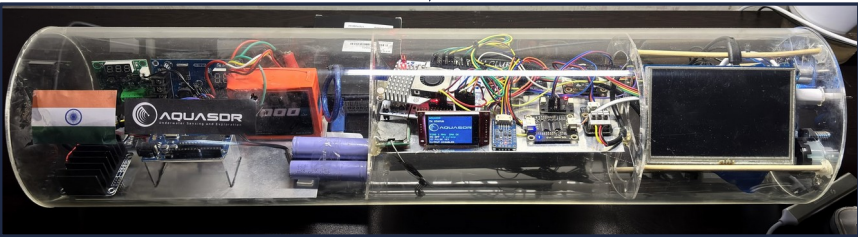
CLEAR → shorter / sharper ping
MURKY → lower band / longer pulse / higher output demand

05 CONDITION • ANALOG FRONT END



2nd-order low-pass filter
70.7 kHz design target
TLV9062 x1.5

DESIGNED • NEXT BUILD



AQUASDR PAYLOAD • PHYSICAL PROTOTYPE acrylic payload hull

External scope validation → PENDING • AUV hull-slot pod → NEXT BUILD

